Practical Assignment: Forensic Image Examination & Metadata Analysis

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1. Student Information

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Course/Subject: Digital Forensics
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2. Evidence / Image Information

The forensic image examined during this practical assignment was created during Task 1. The image parameters and verification hashes obtained directly from FTK Imager are recorded below:

- **Image Filename:** scene.pdf.ad1
- **Image Format / Type:** AccessData Custom Content Image ([Multi]) [AD1]
- **Source Information / Path:** Forensic_Test:C:\Users\HomePC\Desktop\ Forensic_Test\
- **Image Size:** 2,375 Bytes (3 KB logical size)
- **MD5 Verification Hash:** 21ea09027ad80cc9e84fea...
- **SHA1 Verification Hash:** 943f7d231987ece78ec804...

[INSERT SCREENSHOT HERE: Full FTK Imager Window with Loaded Image & Verification Hashes]

Figure 1: FTK Imager interface showing loaded AD1 evidence container, root node structure, and verification hashes in the Properties Pane.

3. Step-by-Step Examination Procedure

The forensic examination was conducted strictly adhering to standard digital forensics guidelines without altering original evidence:

1. **Application Launch:** Launched Exterro FTK Imager (Version 8.3.0.27) on the forensic workstation.
2. **Evidence Acquisition:** Selected *File > Add Evidence Item...*, chose *Image File* as the source type, and navigated to load scene.pdf.ad1.
3. **Interface Verification:** Configured and reset docked windows (*View > Reset Docked Windows*) to ensure the Evidence Tree, File List, Properties Pane, and Viewer Pane were active.
4. **Directory Navigation:** Explored the directory structure under Forensic_Test to analyze logical hierarchy and inspect zero-byte and payload files.
5. **Metadata Extraction:** Recorded MACB timestamps, DOS file attributes, sizes, and path information for five selected objects.
6. **Hexadecimal Examination:** Toggled the Hex Preview mode (HEX button) on scene.pdf.ad1 to analyze raw header/payload bytes.

[INSERT SCREENSHOT HERE: Directory & File Structure View]

Figure 2: Directory navigation within FTK Imager showing file objects listed under the Forensic_Test directory.

4. File Metadata Record

Five distinct objects/files were selected from the forensic image for complete metadata extraction:

Property	File 1	File 2	File 3	File 4	Item 5 (Root Container)
Filename	collect.txt	sample.txt	scene.pdf	scene.pdf.ad1	Custom Content Image([Multi]) [AD1]
File Path / Location	...\Forensic_Test\	...\Forensic_Test\	...\Forensic_Test\	...\Forensic_Test\	Evidence Root Container

Property	File 1	File 2	File 3	File 4	Item 5 (Root Container)
File Type	Regular File	Regular File	Regular File	Regular File	AccessData AD1 Image
File Size	0 Bytes	0 Bytes	0 Bytes	2,375 Bytes (3 KB)	Logical Image Container
Date Created	9/2/2026 6:05:54 PM	9/2/2026 6:05:08 PM	9/2/2026 6:05:54 PM	9/2/2026 6:39:24 PM	9/2/2026 6:39:24 PM
Date Modified	9/2/2026 6:05:54 PM	9/2/2026 6:05:08 PM	9/2/2026 6:06:09 PM	9/2/2026 6:39:24 PM	9/2/2026 6:39:24 PM
Date Accessed	9/2/2026 6:05:54 PM	9/2/2026 6:05:08 PM	9/2/2026 6:06:09 PM	9/2/2026 6:39:24 PM	9/2/2026 6:39:24 PM
Attributes	Archive: True	Archive: True	Archive: True	Archive: True	Evidence Image Object

[INSERT SCREENSHOT HERE: Selected File Properties & Metadata]

Figure 3: FTK Imager Properties pane displaying specific file metadata, timestamps, and attributes for selected items.

5. Potentially Relevant Digital Evidence Findings

1. **scene.pdf.ad1 (Path: ...\Forensic_Test\scene.pdf.ad1)**
   Observation: Size is 2,375 Bytes containing valid hexadecimal structures and raw byte streams.
   Relevance: Serves as the primary active data container within the evidence set. Unlike surrounding 0-byte items, it contains inspectable payload data relevant to the investigation.

2. **scene.pdf (Path: ...\Forensic_Test\scene.pdf)**
   Observation: 0 Bytes in size, created at 6:05:54 PM and modified at 6:06:09 PM.
   Relevance: The 15-second timestamp difference between creation and modification on a 0-byte PDF suggests an interrupted export, failed file creation, or purposeful data wipe/truncation.
3. **sample.txt (Path: ...\Forensic_Test\sample.txt)**
   Observation: 0 Bytes in size, creation timestamp 6:05:08 PM.
   Relevance: Represents the earliest recorded file activity within the target folder, establishing the baseline starting time for staging activity on the system.

[INSERT SCREENSHOT HERE: Hexadecimal Preview Pane Showing Raw Data for scene.pdf.ad1]

Figure 4: Hex Preview Pane displaying raw hexadecimal values and ASCII representation for scene.pdf.ad1.

6. Timestamp Sequence & Forensic Analysis

Direct Observations (Facts)

- The initial file creation recorded in the folder was sample.txt at **6:05:08 PM**.
- collect.txt and scene.pdf were created simultaneously 46 seconds later at **6:05:54 PM**.
- scene.pdf underwent modification 15 seconds after creation, updating its modified timestamp to **6:06:09 PM**.
- The AD1 container scene.pdf.ad1 was created and written at **6:39:24 PM**.

Forensic Inferences

Based on the chronological progression of timestamps, a user or script initiated staging operations in the Forensic_Test directory starting at 6:05:08 PM. Between 6:05:54 PM and 6:06:09 PM, multiple placeholder files were generated, with brief interaction on scene.pdf. Approximately 33 minutes later (6:39:24 PM), the user/system compiled these artifacts into the scene.pdf.ad1 container file.

7. Results & Examination Summary

- **Image Load Status:** The custom content image scene.pdf.ad1 was successfully loaded, parsed, and verified in FTK Imager.

- **Available Information:** Logical file hierarchy, MACB timestamps, file attributes, system paths, cryptographic MD5/SHA1 hashes, and raw hexadecimal streams were fully available.
- **Examined Items:** 5 primary objects examined (collect.txt, sample.txt, scene.pdf, scene.pdf.ad1, and Root Image Object).
- **Key Metadata Observations:** 0-byte placeholder files showed distinct creation/modification shifts within a 1-minute window, contrasting with the 2,375-byte payload file created 33 minutes later.
- **Key Evidence Items Identified:** scene.pdf.ad1 (active container), scene.pdf (truncated/modified placeholder), and sample.txt (earliest activity baseline).
- **Problems / Limitations Encountered:** The original image hash resulted in a mismatch verification warning due to custom content logical imaging constraints, which was documented as a standard forensic limitation. File system volume partition tables were listed as "Not Available" due to the logical nature of the AD1 container.